

Natalia Agapova

ML Engineer

natagapova.ru · Telegram · Email · Google Scholar · GitHub

ML engineer shipping NLP, computer vision, and retrieval-augmented (RAG) systems end to end, from model design and evaluation (metrics, ablations, honest trade-off analysis) to deployed, user-facing interfaces. Comfortable owning a product's technical execution and communicating trade-offs to cross-functional and non-technical stakeholders. Published researcher in model explainability and debiasing.

EXPERIENCE

ML Software Engineer · Croissan Studio

Apr 2025 to present

- Own AI product features end to end at a full-cycle studio, covering model development, interface design, and how results are presented to end users and clients.
- *Letter of recommendation from the studio's CEO* →

Frontend Lead, ML Developer · Asimov Lab

Feb to Sep 2024

- Led the frontend team at an AI test-generation startup while contributing to the ML side, covering the full path from data to shipped UI in close collaboration with the ML and product teams.

ML Engineer · Freelance

2022 to present

- Applied ML work delivered alongside design and web development for independent clients.

SELECTED AI/ML PROJECTS

Fair Resume Screening

NLP · Explainability · Debiasing · Published

- BERT classifier sorting resumes into 9 IT supercategories (HeadHunter data); used Integrated Gradients (Captum) to audit bias and tested 39+ debiasing configurations.
- Baseline accuracy 60.9%; identified that swapping only the applicant's city flipped the model's prediction in 7.7% of pairs, motivating the debiasing work. Published as first author.

Emotion Detection

Computer Vision · On-device Deployment

- CNN classifier on FER2013 (58.7% accuracy, macro-F1 0.569; up to 60.8% with ensembling + TTA), quantized to INT8 and deployed to both Core ML and ONNX Runtime Web at 1.15 ms latency.

Gesture Input

Computer Vision · Real-time Systems

- Real-time webcam cursor control from pinch / ready / click hand gestures using MediaPipe, with PyAutoGUI control and motion smoothing at ~60 FPS.

Personal Knowledge System

RAG · NLP · Python

- Retrieval-augmented assistant that answers strictly from a user's own PDFs, with filename+page citations. Built with chunking, SentenceTransformer embeddings, ChromaDB, and Ollama (Llama 3.2) as the local LLM backend.

SKILLS

ML / AI: Python, PyTorch · LLMs via Ollama · RAG (ChromaDB, SentenceTransformer) · BERT, Captum · Debiasing, counterfactual analysis · MediaPipe, OpenCV

Deployment & tooling: Core ML, ONNX Runtime, model quantization · Git/GitHub

Product-facing: UI/UX design (Figma), front-end development, translating model behavior into usable interfaces

Languages: Russian C2 (native) · English C1+ · French A2 · Korean A1

EDUCATION

Innopolis University · BSc in Applied Artificial Intelligence

2026

PUBLICATIONS

Agapova, N. A.; Lukmanov, R. A. "Improving Fairness in AI-Powered Recruitment: An Interpretable Resume Screening System", 2026. doi:10.66693/mathai.1017

Kholodov, Y. A.; Salloum, H.; Agapova, N. A. "Quantum Annealing Approaches to Breaking RSA Encryption", *Voprosy kiberbezopasnosti*, 2025. doi:10.21681/2311-3456-2025-4-127-133